# YONGHAO TAN

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## EDUCATION

**Ph.D. in Electronic and Computer Engineering** *Sept. 2023 - Present*

**The Hong Kong University of Science and Technology (HKUST)** *Hong Kong, China*

Supervisor: Prof. Tim Kwang-Ting Cheng

**B.E. in Microelectronics** *Sept. 2019 - Jun. 2023*

**Southern University of Science and Technology (SUSTech)** *Shenzhen, Guangdong, China*

Supervisor: Prof. Fengwei An

Overall GPA: 3.77 / 4.0 | Weighted Average: 90.38 | Rank: 11 / 77

## RESEARCH EXPERIENCE

**5nm UCIe-Enabled Multi-Chiplet Generalizable Rendering Processor** *Mar. 2024 - Sept. 2025*

*AI Chip Center for Emerging Smart Systems (ACCESS), Hong Kong, China*

- Architected a 5nm 4-chiplet GeNeRF processor for generalizable rendering to address the heavy external-memory traffic induced by multi-view feature fetching.
- Introduced a UCIe-enabled cross-die unified cache, distributed source-view management, and balance-aware scheduling to maximize source-view reuse while reducing off-chip and cross-die data movement.
- Silicon results reached 91.43 TOPS/W, 55.43 FPS real-time rendering, and 0.29 uJ/pixel through hierarchical sparsity and hybrid NeRF-SR execution in a 45 mm x 45 mm MCM package footprint.

**55nm ReRAM-on-Logic Stacked LLM Accelerator for Speculative Decoding** *Apr. 2024 - Aug. 2025*

*AI Chip Center for Emerging Smart Systems (ACCESS), Hong Kong, China*

- Architected a 55nm edge LLM accelerator whose logic die is stacked with four ReRAM dies via face-to-face bump bonding to support in-stack storage of draft-model codebooks.
- Developed block-clustered weight compression, local-rotation-based outlier-free W4A8 quantization, and adaptive parallel speculative decoding to reduce both target-model EMA and rejected-draft overhead.
- The prototype delivered 14.08 to 135.69 token/s and a 4.46x to 7.17x throughput speedup over a BF16 speculative-decoding baseline on a 55.98 mm^2 logic die.

**28nm CNN-Transformer Accelerator for Semantic Segmentation** *Nov. 2021 - Sept. 2024*

*AI Chip Center for Emerging Smart Systems (ACCESS), Hong Kong, China*

- Architected a 28nm memory-compute-intensity-aware accelerator for high-resolution ConvFormer and SegFormer semantic-segmentation workloads.
- Combined hybrid-attention processing, data-reuse-oriented layer fusion, and cascaded feature-map pruning to reduce attention-side EMA and eliminate redundant KV and weight movement across fused blocks.
- Silicon results achieved 0.22 uJ/token on SegFormer-B0 and up to 52.90 TOPS/W peak efficiency in a 13.93 mm^2 chip.

## PUBLICATIONS

**A 5nm 91.43 TOPS/W 4-Chiplet Generalizable-Rendering-Processor with UCIe-Enabled Cross-Die-Cache and Balance-Aware Progressive Multi-Level Sparsity**
Tan, Y.*, Ma, S.*, Dong, P., Luo, P., Lei, Z., Lu, W., Ying, G., ... & Cheng, K. T.
*2026 IEEE Custom Integrated Circuits Conference (CICC), IEEE.*

**A 14.08-to-135.69Token/s ReRAM-on-Logic Stacked Outlier-Free Large-Language-Model Accelerator with Block-Clustered Weight-Compression and Adaptive Parallel-Speculative-Decoding**
Dong, P., Tan, Y., Liu, X., Luo, P., Liu, Y., Pang, D., Ma, S., ... & Cheng, K. T.
*2026 IEEE International Solid-State Circuits Conference (ISSCC), IEEE.*

**A 28nm 0.22uJ/Token Memory-Compute-Intensity-Aware CNN-Transformer Accelerator with Hybrid-Attention-Based Layer-Fusion and Cascaded Pruning for Semantic-Segmentation**
Dong, P.*, Tan, Y.*, Liu, X., Luo, P., Liu, Y., Liang, L., ... & Cheng, K. T.
*2025 IEEE International Solid-State Circuits Conference (ISSCC), IEEE.*

**Genetic Quantization-Aware Approximation for Non-Linear Operations in Transformers**
Dong, P.*, Tan, Y.*, Zhang, D., Ni, T., Liu, X., Liu, Y., ... & Cheng, K. T.
*2024 61st ACM/IEEE Design Automation Conference (DAC), IEEE.*

**A Reconfigurable Coprocessor for Simultaneous Localization and Mapping Algorithms in FPGA**
Tan, Y.*, Deng, H.*, Sun, M., Zhou, M., Chen, Y., Chen, L., ... & An, F.
*IEEE Transactions on Circuits and Systems II: Express Briefs, 70(1), 286-290, 2022.*

**A Reconfigurable Visual-Inertial Odometry Accelerator with High Area and Energy Efficiency for Autonomous Mobile Robots**
Tan, Y.*, Sun, M.*, Deng, H., Wu, H., Zhou, M., Chen, Y., ... & An, F.
*Sensors, 22(19), 7669, 2022.*

** Authors marked with an asterisk contributed equally to the corresponding work.*

## HONORS AND AWARDS

**Best Teaching Assistant Award, Department of Electronic and Computer Engineering, HKUST** — *Aug. 2025*
**Outstanding Graduate (School Level), SUSTech** — *May 2023*
**First-Class Outstanding Students Scholarship with the highest score** — *Sept. 2022*
**Undergraduate Innovation and Entrepreneurship Training Program** — *Apr. 2022*
**Shenzhen Longsys Electronics Company Award (Top 2% in the School of Microelectronics)** — *Dec. 2021*
**First Prize, 2021 National College Students FPGA Innovation Design Competition (Top 22 out of 1,341 teams)** — *Dec. 2021*
**First Prize, 2021 International Competition of Autonomous Running Robots (1st place out of 34 finalist teams)** — *Oct. 2021*

## FUNDING AND SUPPORT

**Postgraduate Studentship (PGS) Award in HKUST** — *Sept. 2023 - Present*
**Undergraduate Innovation and Entrepreneurship Training Programs (Provincial Level)** — *Apr. 2022*
**Guangdong College Students' Scientific and Technological Innovation (Provincial Level)** — *Jul. 2021*

## SKILLS

Research Interests: Software/Hardware Co-Design, Model Compression, 3D Processing
Programming Languages: C, C++, Java, Python, SystemVerilog, Verilog HDL, VHDL
Professional Software: AutoCAD, Cadence, Design Compiler, IC Compiler II, MATLAB, Multisim, Silvaco
Languages: English (fluent), Mandarin (native), Cantonese (native)